Lab 7: Fuzzing

1. python phone_repair_quote_parser.py on the buggy module SS1

```
C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>python phone_repair_quote_parser.py
==== Running mask_imei ====
Enter an IMEI number (e.g. 35-209900-176148-1): 35-209900-176148-1
Masked IMEI: ***********1481

==== Running parse_parts_order ====
Enter parts and quantities as PART xQTY separated by ; (e.g. screen x1; battery x2): screen x1; battery x2
Parts order: {'screen': 1, 'battery': 2}

==== Running parse_model ====
Enter the phone brand and model (e.g. Samsung Galaxy S23): Samsung Galaxy S23
Brand and model: ('Samsung', 'Galaxy S23')

C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>
```

2. pytest with the tests failing SS2

```
=========================================================== short test summary info ============================================================
FAILED test_phone_repair_quote_parser.py::test_mask_imei_no_crash - AssertionError: mask_imei crashed with: AttributeError("'NoneType' object has no
 attribute 'replace'") for input None
FAILED test_phone_repair_quote_parser.py::test_parse_parts_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_phone_repair_quote_parser.py::test_parse_model_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_phone_repair_quote_parser.py::test_parse_parts_order_returns_positive_ints - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su
b-exceptions)
FAILED test_phone_repair_quote_parser.py::test_parse_model_returns_two_strings - ValueError: not enough values to unpack (expected 2, got 1)
========================================================== 5 failed, 1 passed in 2.58s ===========================================================

C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>
```

3. python phone_repair_quote_parser.py on the fixed module SS3

```
C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>python phone_repair_quote_parser.py
==== Running mask_imei ====
Enter an IMEI number (e.g. 35-209900-176148-1): 35-209900-176148-1
Masked IMEI: ***********1481

==== Running parse_parts_order ====
Enter parts and quantities as PART xQTY separated by ; (e.g. screen x1; battery x2): screen x1; battery x2
Parts order: {'screen': 1, 'battery': 2}

==== Running parse_model ====
Enter the phone brand and model (e.g. Samsung Galaxy S23): Samsung Galaxy S23
Brand and model: ('Samsung', 'Galaxy S23')

C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>
```

4. pytest with all 6 passing on the fixed module SS4

```
C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>python -m pytest test_phone_repair_quote_parser.py
========================= test session starts =========================
platform win32 -- Python 3.13.15, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7
plugins: anyio-4.15.1, hypothesis-6.170.15, cov-7.1.0
collected 6 items

test_phone_repair_quote_parser.py ......                             [100%]

========================== 6 passed in 0.91s ==========================

C:\Users\Rakshitha\Documents\sem5\SE\L6_L7_59_phone_repair_quote\LAB7>
```